

DATA SCIENTIST

ADAM WOODS



Learn more about me at www.adamwoods.dev

SUMMARY

I work with **machine learning and generative AI** to find the best solutions to real-world problems, building knowledge and skills as I go. I have extensive experience conducting research and experiments to help find the most effective solutions and drawing valuable, **actionable insights** to help make future decisions. I believe this process to be very important as many of the latest developments and novel techniques within data and tech aren't effectively **applied in real-world scenarios**. This is where I excel.

I have also developed several projects to implement these findings, **making robust pipelines** and tools that can be used with new dynamic data to create a significant impact for the organisation and **improve the way we work**.

1st Class Honours
in Computer
Science with AI

Northumbria
University

PROJECTS

LLM Redteaming Evaluation Pipeline - 2025

A pipeline to determine if an LLM response contained dangerous or unwanted behaviour. Included **LLM Judge feature extraction** and **XGBoost model** to align with human evaluation. Now used to evaluate real agentic tools built by clients.

Token Optimisation Research - 2026

Leading the research domain into improving LLM prompts for downstream task performance and **cost reduction** using mutation techniques and **token level analysis**. Allows us to apply these findings to real agentic tools.

Payment Classification - 2024

A machine learning classification project using a **neural network** and over **10,000 data points** to determine whether debt collection customers were likely to make a payment, enabling more informed decisions about which customers to contact and helping to **significantly reduce costs**.

CAREER

2024 - Present
Senior Data Scientist - **PwC**

2022 - 2024
Senior Data Scientist - **UKSL**

OTHER WORK

Exoplanet Classification
Light Curve Search
LLM Bluetesting
LLM Agentic & Code
Redteaming
LLM Data Exfiltration
Prompt Optimisation
SLM Safety
Quantum Computing

EXPERIENCE

The projects above demonstrate my technical experience but throughout my career I have also developed many management and logistical skills to help promote the work I do and upskill others with my knowledge. I have **hosted an AI safety hackathon**, where I led a technical challenge for our cybersecurity team. I have also **presented at the Leeds Digital Festival**, an event attended by over **20,000 people**. Here I demonstrated the LLM red-teaming research we have done. Furthermore, I have been a **code reviewer** and tutor to several apprentices as well as non-technical colleagues. This includes guiding them through projects and even teaching them programming from scratch, one of them is now an SQL Developer.

If my experience seems relevant to the role, I encourage you to visit my website as it goes through many more of my projects and experience. www.adamwoods.dev